

Formal and Scalable Adaptive Memory Dynamics

Memory allocation, selective decay, retrieval, and long-horizon learning under resource constraints

Tier 3 - Doctoral Research Agenda

Adaptive Memory Dynamics Research Program Stage One revision - September 2026

Evidence status: Stage One is a reproducible sklearn-digits-8x8 experiment across five independent seeds. Reinforced retention approximately matches the persistent control on ordinary accuracy and Task B adaptation; passive decay reduces accuracy; neither decay mechanism improves Task A retention in the current sequential setup.

Abstract

This doctoral agenda generalizes adaptive forgetting from a small supervised testbed into a theory-and-systems program for long-horizon learning under finite memory and computation. It studies how retention, decay, reinforcement, retrieval, and resource budgets interact across tasks, architectures, and timescales. The program seeks formal and empirical conditions under which selective forgetting is useful, neutral, or harmful, rather than assuming a universal benefit.

1. Core research axes

- Retention dynamics: what should persist, weaken, consolidate, or be relearned?
- Task and distribution dynamics: how do sequence, recurrence, and drift change optimal retention?
- Resource constraints: how do finite memory, compute, and energy budgets alter the value of forgetting?
- Retrieval effects: when does access or rehearsal modify retention priorities?
- Architecture transfer: which effects persist beyond one MLP or benchmark?

2. Formal program

A general memory state can be represented as a time-varying retention process $R(\theta, t \mid a, f, q, B)$, where θ denotes learned representation state, a activation/reinforcement history, f forgetting dynamics, q retrieval/task context, and B resource budget. This is a research scaffold, not a biological law.

3. Empirical program

Scale	Question
Parameter / feature	Which representations decay and why?
Task	How does selective retention affect backward and forward transfer?
Sequence	How sensitive are conclusions to ordering and recurrence?

Scale	Question
Architecture	Do effects transfer across MLPs, CNNs, transformers or agent memory modules?
System	Does forgetting improve long-duration operation under explicit memory/compute budgets?

4. Evidence standards

Claims require replicated seeds, held-out task sequences, benchmark diversity, effect sizes and uncertainty, preregistered primary outcomes where practical, and explicit comparisons with conventional regularization/continual-learning baselines. Biological language must be separated from computational evidence unless cognitive datasets are directly modeled.

5. Long-horizon contribution

A successful program would produce a reproducible map of retention-adaptation-resource tradeoffs, formal definitions that distinguish passive decay from adaptive retention, and validated conditions under which forgetting is computationally useful. A finding that persistent retention dominates in many regimes would remain scientifically informative.